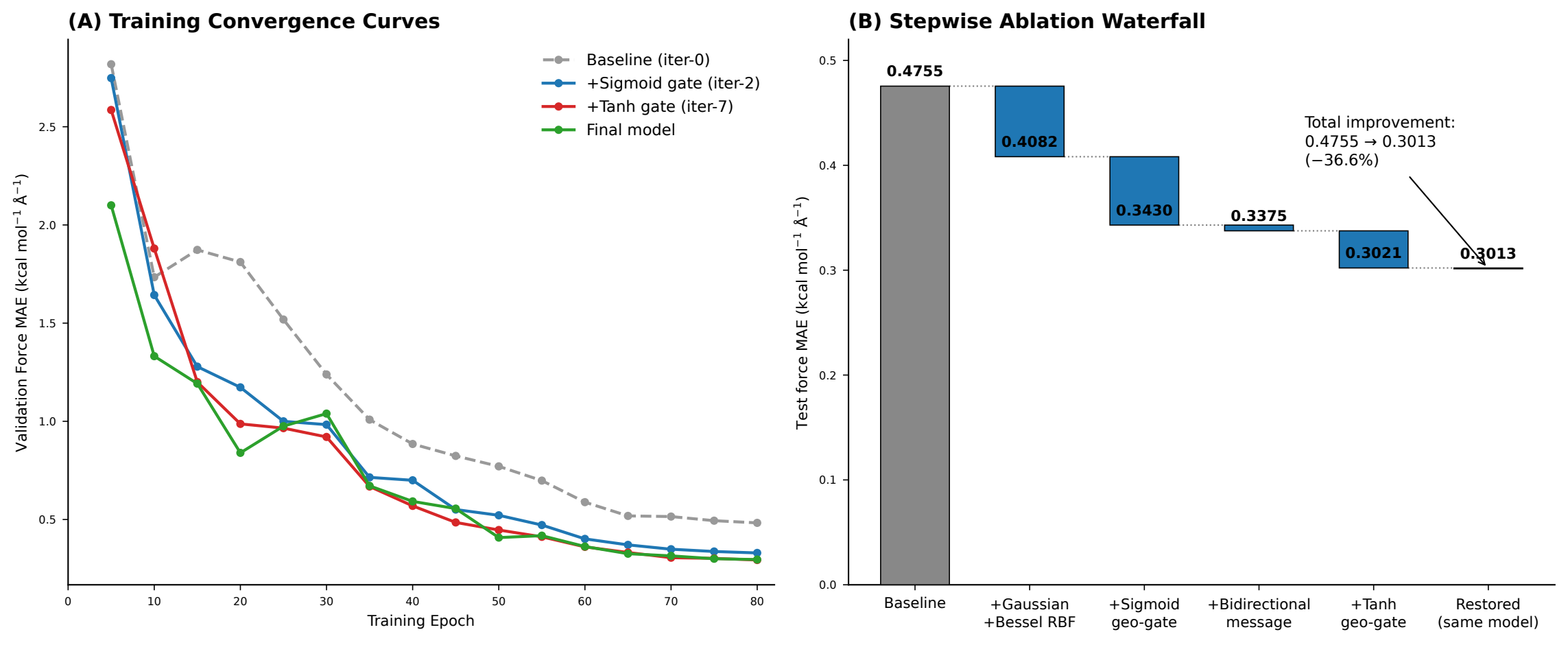
(A) Training Convergence Curves
Baseline (iter-0)
+Sigmoid gate (iter-2)
+Tanh gate (iter-7)
Final model
Validation Force MAE (kcal mol⁻¹ Å⁻¹)
Training Epoch
0
10
20
30
40
50
60
70
80
0.5
1.0
1.5
2.0
2.5
(B) Stepwise Ablation Waterfall
Test force MAE (kcal mol⁻¹ Å⁻¹)
0.0
0.1
0.2
0.3
0.4
0.5
0.4755
0.4082
0.3430
0.3375
0.3021
0.3013
Total improvement:
0.4755 → 0.3013
(−36.6%)
Baseline
+Gaussian
+Bessel RBF
+Sigmoid
geo-gate
+Bidirectional
message
+Tanh
geo-gate
Restored
(same model)